Wavelet-Based Time-Frequency Analysis of Evolving fluids in a Microfluidic Rheometer

Subject:	Meeting with Supervisor
Group name/ identifier	Group 24
Meeting Location, Date & Time	Location: Not recorded Date: 27 August 2026 Time: Not recorded
Group members present	Rohan Mathew, Jianhao Zhang, Xuan Zhang, Ria Mishra, Xingcan Wang (Alex). Supervisor: Saumya.
Apologies	None

1 Actions from previous meetings

Task	**Who is responsible**	**Deadline**
Share the analysis code with the advisor.	Rohan	Not recorded
Review physical criteria for gelation and analyse available rheological data.	Team	Not recorded

2 Items discussed and decisions made

Item	**Decision (if applicable)**
Structural changes and gelation	The existing pipeline could identify structural changes, but a defensible definition of gelation remained unresolved. The supplied papers primarily addressed structural change rather than a labelled gel point.
Soft glassy rheology proposal	Rohan discussed a paper combining a soft glassy rheology model, multi-frequency nonlinear measurements and a neural network. Early prediction reported for clay was a literature result; transfer to blood remained to be investigated.
Code update	The advisor agreed that Rohan could provide updated code by Monday and requested the supporting paper. The relationship between model outputs and observed structural changes should be investigated.
Available data and physical criteria	A team presentation considered storage/loss moduli, loss tangent, frequency independence and phase angle. Of 29 datasets reviewed, 17 were considered usable; most appeared predominantly liquid-like.
Gel point scoring and figures	An exploratory score combined similarity of frequency slopes, loss tangent independence, phase angle agreement and fit reliability. The advisor questioned the interpretation and requested clearer figures and explanation; the score was not established as a validated gel point measure.
Further wavelet work	Work was continuing on applying wavelets to a physically meaningful channel. Further literature review was needed to justify synthetic clotting signals and physical interpretation.

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3 Agreed actions from this meeting

Task	**Who is responsible**	**Deadline**
Send the proposed soft glassy rheology paper to the advisor.	Rohan	Not recorded
Share updated code incorporating the proposed changes where feasible.	Rohan	Monday following the meeting
Revise the gel point score explanation and figure presentation, then send the revision.	Presenting member; name not clear	Not recorded
Continue wavelet analysis on a physically meaningful channel and present progress.	Rohan	Next meeting
Review literature to justify clotting criteria and synthetic signal design.	Team	Not recorded

4 Next meeting

Location	Not recorded
Date & Time	Following week discussed; exact date and time not recorded
Goals	Review revised code, clearer rheological figures and progress on physically meaningful wavelet analysis.

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